SIOP 2026 · ML COMPETITION

ONE HOT KEY

ONE PERSON + AGENTS vs HUMAN TEAMS

Bivariate Pearson r extraction · IO Psychology meta-analysis

Team: One Hot Key · Hidden-set MSE · Walt (RTX PRO 3000 Blackwell)

The Challenge

Build an automated pipeline that reads academic PDFs and extracts **zero-order bivariate Pearson r correlations** for meta-analytic synthesis in industrial-organizational psychology.

Dev set

127 studies on trust × subjective well-being

Test set

23 distinct construct pairs across 66 papers — out-of-distribution

0.013641

BEST DEV SET MSE

127 + 66

STUDIES PROCESSED

23

CONSTRUCT PAIRS (TEST)

SCORED ON **Mean Squared Error vs. hidden ground-truth r values**

SIX GATES, CASCADING FALLBACK

Pipeline Architecture

Each PDF passes through a cascade — earlier gates are cheap and high-precision, later gates are expensive and recover what slips through.

01 **PDF Acquisition** → 02 **Docling Tables** → 03 **Regex Extraction** → 04 **Vision Fallback** → 05 **Opus API Sweep** → 06 **Imputation**

EARLY EXIT: if a high-confidence value is found at any gate, downstream gates are skipped for that pair.

GATE 01 · PIPELINE STAGE

PDF Acquisition

Robust DOI-to-PDF resolution with a four-stage fallback chain

batch_pdf_agent.py walks each study's DOI through progressively more aggressive resolvers, saving as study1.pdf … study127.pdf with a download_log.json audit trail.

- CrossRef → publisher metadata and direct PDF links
- Unpaywall → open-access copies of paywalled articles
- Semantic Scholar → corpus-level fallback when the above miss
- Playwright → headless-browser scrape for HTML-gated PDFs

Playwright

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ACQUISITION RATE 193/193 PDFs successfully retrieved across dev + test sets

Docling Table Extraction

Structured extraction of correlation matrices from academic PDFs

Correlation tables are where the gold lives. Docling parses page layout into structured rows and columns, preserving cell relationships that downstream regex needs.

- Layout-aware: handles multi-column papers and rotated pages
- Preserves cell adjacency so row × column variable lookup works
- Output feeds the regex extractor with normalized table text
- Fails gracefully on scanned PDFs — vision fallback picks up the slack

docling

PYTHON PACKAGE

PRIMARY USE **Correlation matrices, descriptive statistics tables, MBI subscale tables**

Regex Extraction

Pattern-matching the right r value out of dozens of candidates

The deterministic core of the pipeline. Construct-aware patterns walk extracted table text and prose, applying same-wave rules, sign-flip logic for inverse labels, and explicit rejection of false-positive table types.

- Distinguishes Pearson r from Spearman ρ, betas, and probit coefficients
- Rejects ANOVA, imputation, and cross-wave correlation tables
- Handles Unicode significance markers (∗ U+2217, ⁎ U+204E) vs. ASCII *
- Construct-aware sign flips when papers use inverse labels (e.g. distrust)

Python re

PYTHON PACKAGE

PRECISION FOCUS False positives cost as much as misses — explicit reject-lists are critical

Vision Fallback

Visual table parsing for cases where layout extraction fails

When Docling and regex both come up empty — typically scanned PDFs, image-embedded tables, or unusual layouts — the pipeline falls back to a local vision-language model running through Ollama.

- qwen2.5vl reads the page image directly and returns structured cells
- phi4 acts as a text-only re-parser for lightly scrambled extractions
- Runs entirely local — no API cost, no rate limits, full offline reproducibility
- Construct classification and sign-flip logic must be applied post-hoc

Ollama · qwen2.5vl

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WHEN IT FIRES Scanned PDFs, image-only tables, papers Docling parses but regex misses

GATE 05 · PIPELINE STAGE

Structured LLM Extraction

How the hardest cases get extracted — process, not vendor

A separate module — pipeline_v9_opus_sweep.py — runs a custom extraction loop on a curated short-list of studies the open-source pipeline missed or low-confidenced. The model behind the call is interchangeable; the methodology is what makes the values trustworthy.

- PROMPT: construct definitions, sign-flip rules, same-wave longitudinal rule, and explicit rejection criteria for ANOVA / regression beta / cross-wave traps
- INPUT: full PDF + a list of which (X, Y) construct pair to extract for that paper, plus measure aliases pulled from each paper's methods section
- OUTPUT: JSON-schema-constrained — { r, n, page, table_cell, confidence, supporting_quote } — parsed and type-validated before acceptance
- AUDIT: every accepted value carries its supporting quote, so the manifest can be spot-checked by re-reading the cited sentence rather than re-running the model

Schema-constrained sweep

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WHY THIS GATE EXISTS **To recover the ~10% of correlations that survive bad layout, sign-flips, or table-cell ambiguity that defeats every upstream gate**

Imputation

Last-resort fill for studies where every gate above missed

When all five extraction gates return nothing, the pipeline imputes a single value chosen to minimize MSE under uncertainty. The optimal value is back-calculated from paired dev-set submissions.

- Optimal imputation value ≈ 0.152 (data-driven, not a heuristic)
- Back-calculated by submitting paired dev runs and solving for the minimum
- U-curve flat at optimum — further imputation tuning yields no gain
- Coverage at upstream gates is the only remaining lever for improvement

pandas · numpy

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KEY INSIGHT **Imputation is not a differentiator. Pipeline coverage is.**

Division of Labor

Matt

Strategic lead

Pipeline strategy, GT coding, PDF reading, batch-log diagnosis, prompt-writing for Cursor.

Claude

Diagnostician

PDF inspection, batch failure analysis, precise Cursor-prompt drafts, file-validation scripts.

Cursor

Implementer

All code reading, concrete bug-finding, and pipeline implementation. No architectural decisions made unilaterally.

PowerShell

Runtime

Batch runs and spot checks only — never used as an iterative debugging surface.

GOLDEN RULE Diagnose once, write one comprehensive prompt, let Cursor implement. Two cycles maximum.

DEV → TEST OUT-OF-DISTRIBUTION

Validation Strategy

No traditional k-fold CV — the competition provides a fixed dev/test split with deliberate distribution shift in construct space.

DEV SET

127

studies on trust × subjective well-being

- Held-out ground-truth manifest manually coded
- 57 numeric, 8 confirmed BLANK, 1 PENDING
- MSE used as fast iteration signal

TEST SET

66

papers across 23 distinct construct pairs

- Generalizes well beyond trust × wellbeing
- Forces removal of all study-specific hardcodes
- Hidden-set MSE — only visible after submission

PAIRED-SUBMISSION TRICK

Optimal imputation value back-calculated by submitting matched dev runs and solving the resulting two-equation system. The U-curve was confirmed flat at the optimum, ruling out further imputation tuning as a path forward.

DEV vs. TEST

Results

DEV SET

0.0136

Best MSE · submission_v11_study59

Held-out trust × wellbeing manifest
Used as fast iteration signal

TEST SET

0.0351

Best test MSE · 2026-04-11

23 construct pairs across 66 papers
Out-of-distribution generalization

WHAT CLOSED THE GAP

Removing study-specific hardcodes from v11, generalizing construct classification to 23 pairs, and recovering false-negative regex matches on high-r studies (study41 = 0.73, study17 = 0.65, study56 = −0.57).

Dev Set Leaderboard

How a one-person-plus-agents experiment landed against teams of researchers and analysts on the dev set.

RANK	TEAM	ERROR ↓	SUBMITTED
1	Hungry Llama	0.006336	2026-04-06
2	MetaML	0.009370	2026-03-25
3	Putka's Army	0.011521	2026-03-29
4	ILMNinsights	0.013198	2026-03-26
5	STATS Lab at CMC	0.013510	2026-04-01
6	***One Hot Key***	**0.013641**	2026-04-05
7	DSTST	0.015793	2026-04-08
8	Cam the Ram	0.026518	2026-03-23
9	Chen-Chillas	0.029143	2026-04-10
10	Neural Nomads	0.054047	2026-04-08

6 / 10

Dev set rank

THE ASYMMETRY

1 developer + Claude + Cursor.

Within 0.0005 MSE of the lab and the squad ranked above. Test set ranking pending close of competition.

Limitations

WORKSTATION · Walt

Core Ultra 9 285HX · 24 cores (8P+16E) / 24 threads · **RTX PRO 3000 Blackwell** · 12 GB GDDR7 · **64 GB DDR5**

12 GB VRAM forces phi4 over the bigger model

qwen2.5:72b — Ollama's strongest open-weights extraction tier — needs ~40 GB VRAM at Q4, over 3× what the RTX PRO 3000 Blackwell carries. Phi4 (~14B, ~9 GB Q4) is the largest model that stays fully GPU-resident; the 72B class would spill to CPU and lose roughly an order of magnitude in throughput.

Single-machine sequential

All 193 PDFs run through six gates on one workstation. Cluster-scale competitors run dozens of pipeline variants in parallel; we run one full sweep at a time, which gates how many ideas we can test per day.

Ollama serializes inference

24 cores and 992 AI TOPS, but Ollama runs one vision query at a time per model. Effective throughput on Gate 4 is bounded by single-stream latency, not total compute — so even the model that fits is queue-bound on a 127-paper batch.

Vision gate capability ceiling

qwen2.5vl-7B fits the 12 GB budget, but the 32B-and-up vision tier that would close the scanned-table OCR gap doesn't fit without aggressive quantization that erodes the accuracy gain. Vision is the most accuracy-sensitive gate in the pipeline, and it's structurally capped one tier below state-of-the-art.

WHAT THIS PROJECT TAUGHT US

Key Learnings

No study-specific hardcodes

Every fix must generalize across the 23 construct pairs. Study-specific overrides were identified and removed during the v11 → v12 migration.

Silent failures are the hardest bugs

dynamic_mode never fired because process_study skipped build_study_config; Unicode asterisks vs ASCII broke parsing in study59. Both required targeted internal-state dumps to surface.

False positives cost as much as misses

ANOVA tables, imputation tables, Spearman correlations, and cross-wave pairs all required explicit rejection logic. Precision is not free.

Coverage, not imputation, is the lever

Optimal imputation was back-solved analytically. The U-curve is flat. The only remaining path to a lower MSE is recovering the missed high-r studies upstream.

Thank you · Questions?